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Journal of Adhesion Science and Technology

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/tast20>

Preparation and characteristics of polypropylene-graft-maleic anhydride anchored micro-fibriled cellulose: its composites with polypropylene

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Published online: 19 Nov 2014.



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To cite this article: Jun-Seok Yeo & Seok-Ho Hwang (2015) Preparation and characteristics of polypropylene-graft-maleic anhydride anchored micro-fibriled cellulose: its composites with polypropylene, Journal of Adhesion Science and Technology, 29:3, 185-194, DOI: [10.1080/01694243.2014.980632](https://doi.org/10.1080/01694243.2014.980632)

To link to this article: <http://dx.doi.org/10.1080/01694243.2014.980632>

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Preparation and characteristics of polypropylene-graft-maleic anhydride anchored micro-fibriled cellulose: its composites with polypropylene

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(Received 27 July 2014; final version received 22 September 2014; accepted 21 October 2014)

The surface modification of micro-fibriled cellulose (MFC) was carried out through the hydrolysis–condensation reaction using (3-aminopropyl)triethoxysilane (APS) and then was subsequently treated with polypropylene-graft-maleic anhydride (MAPP) to give desired MAPP-anchored MFC (MAPP-a-MFC). The confirmation of surface modification for the pristine MFCs was characterized by FTIR, EDX, and XPS measurements. The polypropylene (PP) composites with MAPP-a-MFC were prepared via typical melt-mixing method. The effects of MAPP-a-MFC on the physical properties of the PP composites were investigated. From the results of mechanical properties, thermal behaviors, and the morphology of the composites, it was found that there was an improved interfacial adhesion between the MAPP-a-MFC and PP matrix in the composites.

Keywords: micro-fibriled cellulose; polypropylene; polypropylene-graft-maleic anhydride; interfacial adhesion; composite

1. Introduction

Cellulose, as a highly popular polymeric material, has many utilitarian benefits such as renewability, biodegradability, non-toxicity, and so on. During the past decade, tremendous attention has been paid to the application of cellulosic materials as novel composite components [1–6] because the cellulose-reinforced polymeric composites can serve many advantages such as low production cost, low density, high modulus, fast biodegradation, and good mechanical properties. Among them, cellulose has been attracted much attention as a filler material for polypropylene (PP) composites, recently.[7–12] Unfortunately, due to the inherent incompatibility between hydrophilic cellulose fibers and hydrophobic synthetic polyolefin, poor interfacial adhesion which results in impaired properties to the composites occurs. Because the properties of such composites depend on the degree of interfacial adhesion as well as the properties of the individual components, the modification of pristine cellulose might play a key role in the cellulosic polymer composites.[13–15]

Numerous research groups have been performed to enhance the interfacial adhesion for efficient transfer of stress from the matrix to the fibers, using physical treatments [16–18] as well as chemical treatments [1,19–23] (pretreatment of fiber surfaces by coupling agents, such as silanes and isocyanates, and/or modification of the matrix by

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grafting with reactive moieties, such as acrylic acid, acrylic esters, maleic anhydride, etc.). Of all these attempts, maleic anhydride grafted polyolefins were found to be one of the most efficient for a composite of cellulose fibers and polyolefin matrices.[20–23]

This work describes a new approach for surface modification of a pristine micro-fibriled cellulose (MFC) through hydrolysis–condensation reaction of a triethoxysilane coupling agent, (3-aminopropyl)triethoxysilane, followed by amidation of amine groups with polypropylene-*graft*-maleic anhydride (MAPP) to produce an polypropylene-*graft*-maleic anhydride anchored micro-fibriled cellulose (MAPP-a-MFC). This strategy will increase interfacial adhesions between the MFCs and the PP matrix which ensure the good mechanical properties for the composite.

2. Experimental section

2.1. Materials

The isotactic polypropylene (iPP) pellets (trade name; SEETEC H1500, melt flow index (MFI); 12 g/10 min at 230 °C, density; 0.90 g/cm³) was obtained commercially from LG Chemicals (Korea). Fibrous cellulose (CF11; 50–350 mm in length, about 20 mm in diameter, I type cellulose with a crystallinity of 93%), (3-aminopropyl)triethoxysilane (APS), and MAPP pellets (maleic anhydride content: 8–10 wt.-%, Mw ~9100 g/mol) were purchased from Aldrich. Dehydrated acetone and dehydrated xylene were purchased from Sanchun Chemicals, Korea.

2.2. Cellulose treatments

A solution of APS was prepared by adding 30 g of APS to 500 mL of a methanol–water mixture (9/1; v/v) and stirred for 30 min. A 10 g of pristine MFCs was added to the APS solution and stirred for 3 h. The silane-treated MFC were filtered and washed with pure water and finally dried *in vacuo* at 60 °C. In next step, MAPP was dissolved in xylene at 120 °C. When it was completely dissolved, the silane-treated MFC was added in the xylene solution and stirred for 6 h at 120 °C. The product was hot-filtered and washed with hot-xylene solvent several times, then kept *in vacuo* at 80 °C for 3 days to evaporate the solvent completely.

2.3. Preparation of PP composites

The MAPP-a-MFC/PP and pristine MFC/PP composites were prepared in the Haake internal mixer (Rheomix 600p). The mass ratios of cellulosic filler to PP were 5/95, 10/90, 15/85, 20/80, 25/75, and 30/70. The melt-mixing was carried out at 200 °C and 55 rpm for 10 min. Sheets of about 1 mm thickness were obtained by compression-molding in a Shinto press (Shinto Metal Ind. Ltd., Japan) at 200 °C. Imide films were used to avoid the adhesion of PP to the stainless surface of the mold. Standard specimens were cut from the pressured sheet for measurement of tensile properties.

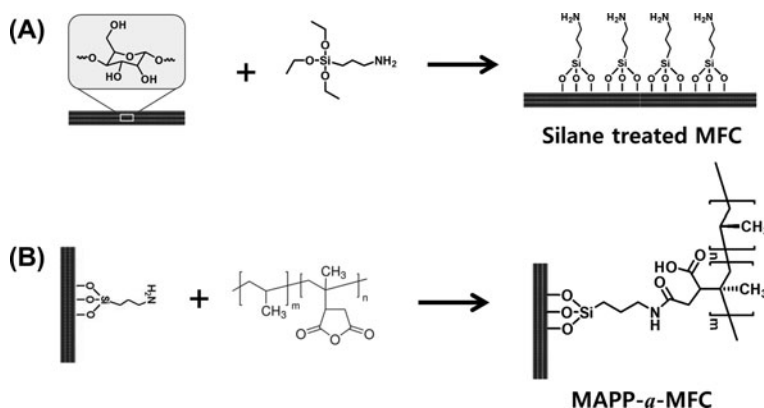
2.4. Equipment and experiments

The thermal behavior of the phase transitions was studied using a Perkin Elmer PYRIS Diamond DSC with an Intracooler 2P apparatus. Tensile properties were measured with a Shimadzu AGS-5kNG universal testing machine in accordance with the testing method

for tensile properties of plastics (at a strain speed of 5 mm/min). The morphology of the PP composites was examined using a JSM-7100F field emission scanning electron microscope (FE-SEM, Jeol, Japan) at 10 kV. The chemical composition of the treated MFCs was analyzed by using energy dispersive X-ray (EDX, Oxford Instrument) spectrometer attached to the FE-SEM. The XPS spectra were acquired on a K-Alpha (Thermo Scientific Co.), using a monochromatic Al K α source operating at 72 W.

3. Results and discussion

Surface chemical modification of the MFCs having the high polar character of MFC surface will improve their dispersion ability in hydrophobic polymer matrices. In this study, the surface modification of the pristine MFC was performed by two steps including hydrolysis–condensation and amidation reaction. In the hydrolysis–condensation reaction, as first step, it is well-known from the literature [24] that the hydrolyzable alkoxy group in the presence of moisture leads to the formation of silanols. This silanol reacts with the hydroxyl group on the MFC surface to form stable covalent bonds on the MFC surface.[24] This expected reaction between pristine MFC surface and APS is illustrated in Scheme 1(A). To better understand the hydrolysis condensation reaction between APS and hydroxyl group on MFC surface, SEM/EDX was used to confirm the presence of silicon elements on silane-treated MFC indicating the retention of the siloxane. Figure 1 shows the EDX spectrum and SEM microphotograph of the silane-treated MFC surface. Figure 1(C) presents the EDX spectrum of the silane-treated MFCs. The intensity for the silicon peak at 1.75 keV [25] as regarding the C and O K α emission peaks was a quite small and gold element peak coming from Au-coating was also detected. Despite the small intensity of the silicon element, it does give evidence about the occurrence of the grafting with the silanes. Figure 1(B) shows silicon element mapping image obtained from the magnified SEM image (Figure 1(A)). In Figure 1(B), the variation in brightness over the background represented the localized distribution of silicon element across the MFC surface. The brightness in the images indicated retention of silicon elements. These results should suggest that the silane moieties are distributed evenly on the surface of the silane-treated MFCs.



Scheme 1. Preparation of MAPP-a-MFC via a hydrolysis condensation (A) and amidation reaction (B).

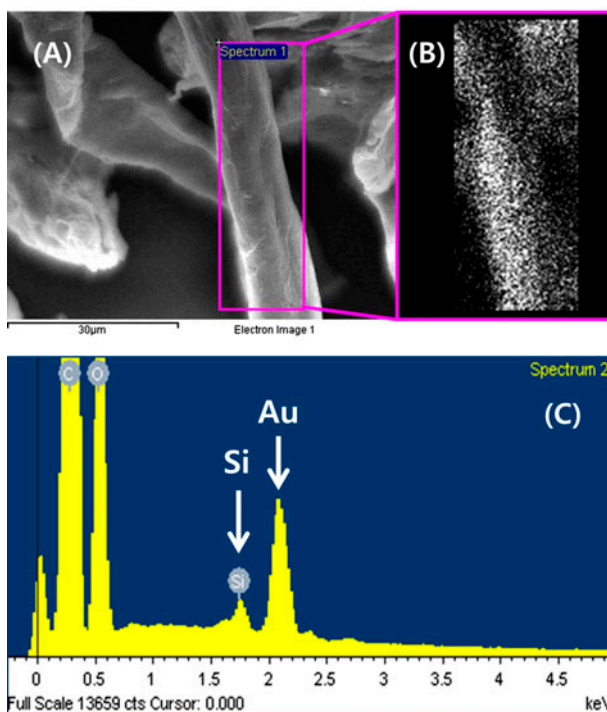


Figure 1. SEM micrograph (A), silicon element mapping image (B), and EDX spectrum of the silane-treated MFCs (C).

Previous works have been demonstrated that MAPP is one of the good compatibilizer for cellulose and PP composites [9,26] due to the ability of esterification reaction between MAPP and hydroxyl group of cellulose during the melt-mixing process. For the highly crystalline cellulose, the ester bonding between MAPP and cellulose is too few to play a role as a compatibilizer in the polymeric composites, unfortunately. They have a very limited amount of free hydroxyl groups on surface of the highly crystalline cellulose due to the intra- and intermolecular hydrogen bonding characteristics. Furthermore, the large steric hindrance of the maleic anhydride grafts should restrict the proximity of the maleic anhydride groups to the cellulose surface.[27] Thus, in order to attain a cellulose bearing a plenty of compatibilizer materials, the MAPP-a-MFC was prepared by surface amidation of the silane-treated MFC with MAPP under boiling xylene solution (Scheme 1(B)). We assume that treatment of the silane-treated MFCs with MAPP can give a desired MAPP-a-MFC coupled between a maleic anhydride and primary amine end group of the silane-treated MFC. The success of this reaction was confirmed by fourier transform infrared spectroscopy (FTIR) as well as X-ray photoelectron spectroscopy (XPS). FTIR peaks corresponding to the molecular vibrations of the MAPP-a-MFC and pristine MFC are shown in Figure 2. When compared to the spectra, a typical FTIR spectrum of MAPP-a-MFC, as that shown in Figure 2, indicated the appearance of a carbonyl amide stretch at 1713 cm^{-1} , C–N stretch at 1376 cm^{-1} , and N–H stretch (hydrogen bonded) at 2950 and 2920 cm^{-1} , which confirmed covalent attachment of the MAPP on the silane-treated MFC surface.

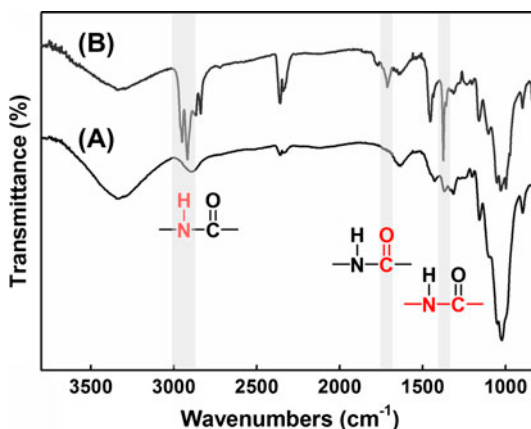


Figure 2. FTIR spectra of the pristine MFC (A) and the MAPP-a-MFC (B).

XPS analysis was also carried out to ascertain more precisely the chemical composition of the MAPP-a-MFC surface because it is very useful for characterizing the quantitative elemental composition and functional groups of the surface of a substance to a depth of about 1–10 Å.[28] At low resolution of XPS spectra, three distinct C_{1s} , O_{1s} , and N_{1s} peaks were detected at 285.0, 531.8, and 398.4 eV, respectively. The high-resolution deconvolutions of the peak corresponding to C_{1s} and N_{1s} in XPS spectrum of the MAPP-a-MFC were given in Figure 3. The analysis of the total C_{1s} high-resolution deconvolution of the MAPP-a-MFC showed that they consist of four component peaks at around 284.3, 285.4, 287.3, and 288.8 eV arising from C1 ($C-C-C$), C2 ($C-O-C$), C3 ($NH-C=O$), and C4 ($COOH$), respectively. A new peak of MAPP-a-MFC (Figure 3(A)) at 287.7 eV,[29,30] attributed to the $O=C-NH-$ amide group revealed that this evidences the amide linkage between MAPP and silane-treated MFC. The peak for the amide linkage was about 1.5 eV less than the carbonyl peak ($C=O$) in carboxylic ($-O-C=O$) acid group, because of an electron donation from the adjacent nitrogen atom. The nitrogen in the amide contains non-bonding electrons that are in a P-type orbital; oxygen in the carbonyl group has electrons in the same orbital. These P-type electrons can conjugate, allowing the electrons to ‘smear out’ along the C–N bond. The chemical effect of conjugation is a stiffer C–N bond than in the amine group, where an electron donation from the adjacent nitrogen atom may decrease the C–N binding energy. This stiffer bonding in the amide results in a higher C–N binding energy.[31] The N_{1s} XPS spectrum fitting (Figure 3(B)) shows the presence of two different nitrogen-based functions: amines (N1; at 398.6 eV) and amides (N2; at 400.8 eV).[32] Interestingly, the majority of nitrogenated functions was amides. This result is another evidence of an amide linkage formation between the silane-treated MFC and MAPP. Consistent with the FTIR results, these XPS results confirmed the construction of chemical linkage between MAPP and silane-treated MFC.

SEM was used to characterize the morphology of PP composites with MAPP-a-MFC and pristine MFC as control. Figure 4 shows typical SEM micrograph of freshly fractured surface for the PP composites filled with MAPP-a-MFC and pristine MFC, respectively. The surface of the PP composites with MAPP-a-MFC was smoother than that obtained from the PP composites with pristine MFC. For PP composites with MAPP-a-MFC, the rupture was clean and the MAPP-a-MFC particle was found to be

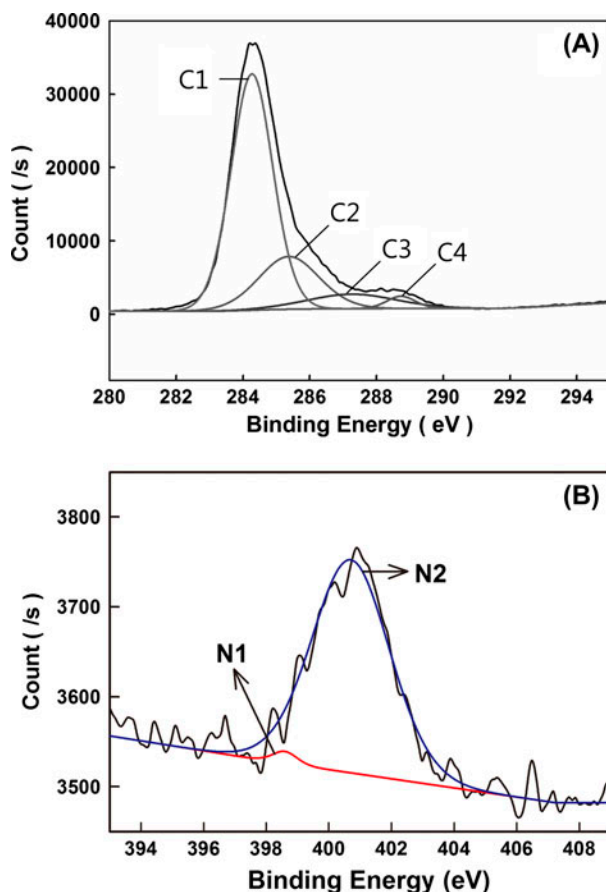


Figure 3. Detailed XPS deconvoluted scans of C_{1s} core level (A) and N_{1s} core level (B) for the MAPP-a-MFC.

distributed evenly throughout the PP matrix (Figure 4(A)). In addition, the magnified SEM micrograph of PP composites with MAPP-a-MFC shown in Figure 4(B) clearly show that all MAPP-a-MFCs were tightly connected with the PP matrix like as them seems to be the continuous phase. By contrast, for the PP composites with pristine MFC (Figure 4(C)), show that there were large gaps and voids in the PP matrix. These results clearly demonstrated that the MAPP-a-MFC provides enhanced interfacial adhesion and good wetting with the PP matrix. In the binary composite system, the improved interfacial adhesion is generally consistent with the increased mechanical properties.[9] Thus, in this system, we can also expect the enhanced mechanical properties, indicating that MAPP anchored on the MFC surface acts as a compatibilizer between pristine MFC and PP matrix.

The melting behavior of the PP composites with MAPP-a-MFC were almost the same as that of neat PP, and not dependent on the MAPP-a-MFC content, as shown in Figure 5. However, the crystallization behaviors of the composites were certainly different, depending on MAPP-a-MFC content. The peak temperatures of crystallization (T_c), for the PP composites with MAPP-a-MFC are plotted against the MAPP-a-MFC content in Figure 5. The T_c values of the PP composites increased with increasing

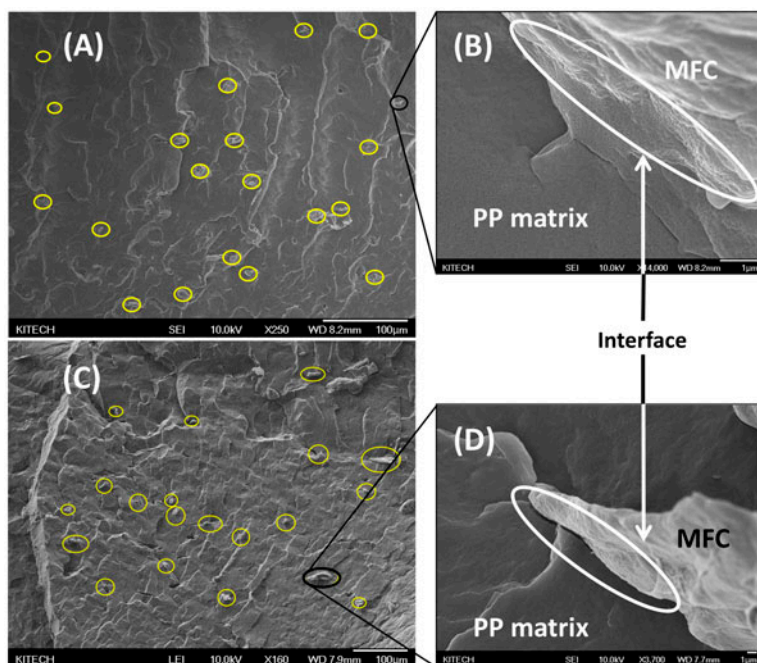


Figure 4. SEM micrograph of cryogenically fracture surface and magnified SEM micrographs: (A) the PP composite containing 10 wt.-% of MAPP-a-MFC, (B) magnified SEM micrograph of (A), (C) the PP composites containing 10 wt.-% of pristine MFC, (D) magnified SEM micrograph of (C) (circles in the figure indicate the MAPP-a-MFC in PP matrix).

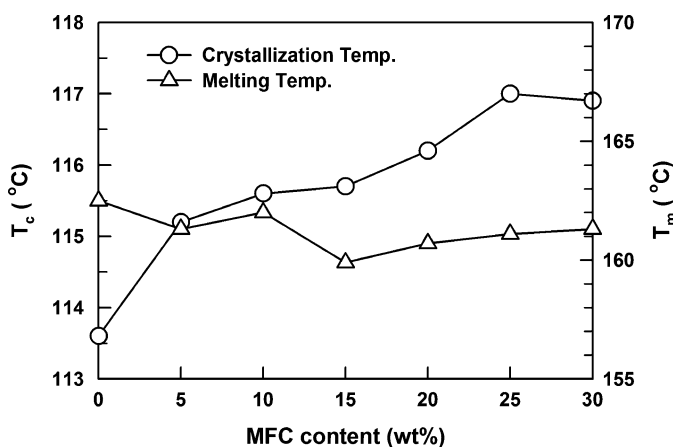


Figure 5. Effect of MAPP-a-MFC content on crystallization temperature (○) and melting temperature (△) of polypropylene.

MAPP-a-MFC content. MAPP-a-MFC particles can act as a nucleating agent for the crystallization of PP matrix, and consequently increases the crystallization temperature as well as the crystallization rate of PP matrix. Such enhanced nucleation may result from the good interfacial adhesion between the MAPP-a-MFC and PP matrix.[9,11,33]

Tensile strength and tensile modulus of the composites with MAPP-a-MFC and pristine MFC are shown in Figure 6. The tensile strength of the composites with pristine MFC did not show a characteristic change but those of composites with MAPP-a-MFC was significantly increased with increasing cellulosic filler content due to the enhanced interfacial adhesion of the composites by MAPP, which was anchored on MFC surface.[34–36] Obviously, an increasing amount of pristine MFC decreases the tensile modulus of the composites due to the weak interfacial adhesion and low compatibility between the pristine MFC and hydrophobic PP matrix.[37,38] While the drastic increase in tensile modulus for the composites with MAPP-a-MFC is undoubtedly caused by the improved interfacial adhesion as well as good dispersion of the treated MFC particles in the PP matrix. The enhanced interfacial adhesion and the treated MFC dispersion can confer the effective transfer of stresses through the interface, resulting in much greater mechanical strength for the PP composite with MAPP-a-MFC.

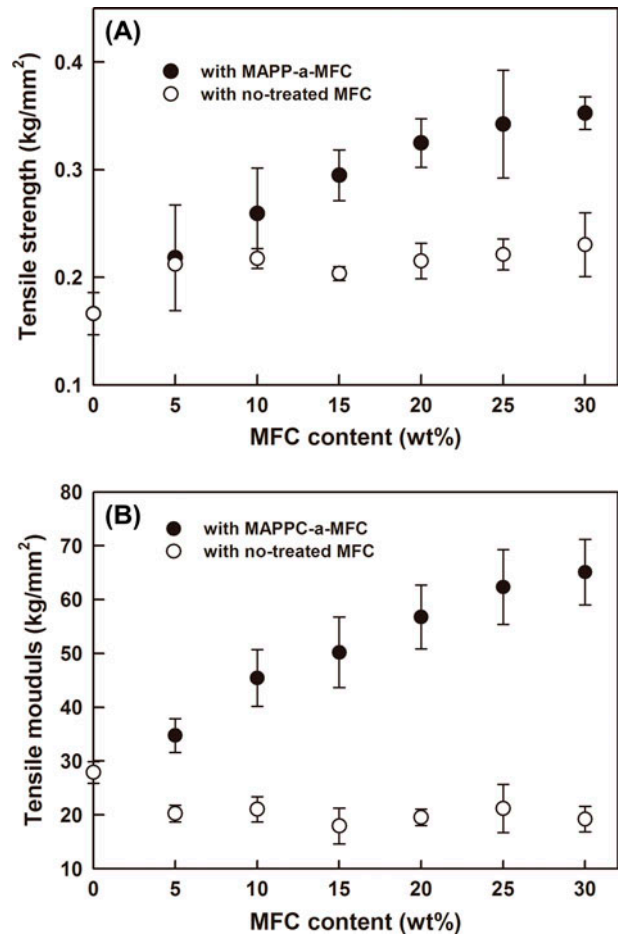


Figure 6. Effect of MAPP-a-MFC and pristine MFC content on the tensile strength (A) and tensile modulus (B) of the cellulosic PP composites.

4. Conclusions

The construction of chemical bonding on MFC surface with MAPP was carried out through amidation reaction between a maleic anhydride and primary amine group of the silane-treated MFC, successfully. PP composites with MAPP-a-MFC were prepared under melt-mixing. Great number of MAPP chains bind to the MFC particles to form a MAPP-a-MFC, creating not only stronger interfacial adhesion between the treated MFC particles and the PP matrix but also greatly improved dispersion of the treated MFC particles in the PP matrix, which in turn results in the greater tensile strength of the PP/cellulose composites.

Acknowledgment

Authors thank to the financial support from GRRC program of Gyeonggi Province (GRRC Dankook 2014-B01).

Funding

This work was supported by the Industrial Strategic Technology Development program (Project No. 10041173) funded by the MOTIE.

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